

Investigating the Effectiveness of Random Forests in Detecting Credit Card Fraud

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Abstract—Credit card fraud detection presents a significant challenge in the financial sector, with global losses reaching \$32 billion in 2021. The difficulty in identifying fraudulent transactions stems from the extreme class imbalance in fraud datasets, where fraudulent cases often account for less than 1% of total transactions, causing traditional machine learning models to favour the majority class and underperform. Random Forests were selected for this study due to their robustness, interpretability, and effectiveness in managing high-dimensional data and class imbalance. The research focused on three key areas: the impact of data balancing techniques, the role of model refinement, and the replicability of previous research outcomes. Data balancing methods, including SMOTE, undersampling, and class weighting, were assessed for their influence on recall and overall performance. Model refinement techniques, such as boosting and bagging, were considered for improved detection, with hyperparameter tuning further optimising performance. The results showed that Feature reduction based on importance metrics provided the best balance between precision and recall, while boosting techniques, particularly XGBoost, significantly enhanced fraud recall and F1 scores. The findings suggest that customised data balancing and refinement strategies can significantly enhance Random Forest performance in fraud detection, providing valuable insights for deploying machine learning models in real-world financial systems.

Index Terms—Random Forest, Fraud Detection, Credit Card

I. INTRODUCTION

A. Background and Problem Statement

Credit card fraud is a major global issue, presenting significant economic and operational challenges to the financial sector. With the growth of digital payment systems, fraudulent activities have become increasingly sophisticated and widespread. In 2021, global losses from credit card fraud were estimated at \$32 billion, underscoring the urgent need for effective fraud detection systems to protect both businesses and consumers [1]. These losses not only lead to direct financial implications but also erode consumer trust and significantly increase operational costs associated with fraud prevention and investigation. Developing reliable fraud detection models is critical in addressing these challenges. Traditional methods, such as rule-based systems and manual

reviews, have proven inadequate in keeping pace with the dynamic and evolving nature of fraudulent tactics [2]. Machine learning models, leveraging large datasets, have emerged as powerful tools for automating and scaling fraud detection. Among these, Random Forests are particularly valued for their robustness, ability to handle high-dimensional data, and effectiveness in managing overfitting [3]. Furthermore, their capacity to address the unique challenges posed by highly imbalanced datasets and their interpretability make them a strong candidate for real-world financial fraud detection systems.

B. Objectives and Research Questions

The primary goal of this study is to evaluate the effectiveness of Random Forest models in credit card fraud detection and improve their accuracy through advanced data preprocessing and model refinement techniques. This research addresses three core questions:

- 1) How does data balancing influence Random Forest performance?
- 2) How do model refinement techniques (e.g., boosting, bagging) enhance accuracy?
- 3) Can existing Random Forest-based fraud detection approaches be successfully replicated in this context?

II. LITERATURE REVIEW

A. Existing Methods for Fraud Detection

Fraud detection has been a critical application of machine learning, with several algorithms employed to analyse transactional data and identify suspicious patterns. Common methods include decision trees, support vector machines (SVMs), and ensemble learning techniques like Random Forests. Each approach has strengths and limitations, making them suitable for different aspects of fraud detection.

Decision Trees are valued for their simplicity and interpretability, making them a popular choice in fraud detection. Previous works found that decision trees can outperform more complex models, such as SVMs, particularly in imbalanced datasets where stratified sampling maintains class ratios [4]. However, they are prone to overfitting, limiting their generalisability in larger, noisier datasets.

SVMs excel in high-dimensional spaces by seeking an optimal hyperplane for class separation. Despite their theoretical advantages, SVMs often underperform in practical fraud detection applications due to scalability issues and poor performance with imbalanced datasets. For example, Sahin and Duman found that SVMs struggled to match decision trees' performance in such cases [4].

Random Forests have emerged as a leading method in fraud detection, surpassing both decision trees and SVMs in key areas. As an ensemble technique, Random Forests combine the predictions of multiple decision trees trained on bootstrapped subsets, reducing overfitting and improving generalisation. Prior studies demonstrate the robustness of Random Forests, particularly in handling imbalanced datasets and identifying critical variables for model performance [5], [6].

Additionally, Random Forests benefit from their ability to evaluate feature importance. The utility of metrics such as mean decrease in accuracy and GINI coefficient in identifying key features like capital structure and asset composition, to improve model accuracy has been highlighted [6]. This ability to leverage features without overfitting provides a significant advantage over standalone decision trees.

B. Challenges in Fraud Detection

a) *Imbalanced Datasets and High False-Negative Rates:* The intrinsic imbalance in fraud detection data, such as in credit card transactions, which normally contains less than 1% fraudulent cases, is one of the main challenges [5]. It biases the machine learning models more often toward the majority class, hence giving a high precision for the minority class with poor recall often skewing performance metrics and, hence, a greater risk of unchecked false negatives. Prior research found that feature selection is a must in this regard and that using important variables will enhance the sensitivity to fraudulent transactions [6].

Synthetic sampling, can be used to balance datasets and account for such challenges. Recent techniques include over-sampling through SMOTE for balancing datasets synthetically, promise improved recall and F1-score by generating more samples of the minority class to train with [7], [8]. However, these could also exhibit particular drawbacks: the use of synthetic data is likely to lead either to overfitting or degraded generalisation ability.

b) *Limitations of Traditional Machine Learning Models in Fraud Detection:* Despite the limitations faced by traditional machine learning models such as decision trees or support vector machines; which are inherently limited in their handling of fraud detection challenges, these still possess useful properties such as, interpretability, and good affinity for small datasets [8].

Although effective in high-dimensional feature spaces, SVMs still suffer from highly imbalanced datasets [4]. Reliance on an optimal hyperplane for classification makes them less effective in cases where the minority class is underrepresented, hence yielding poor recall and limited generalisability across varying datasets.

On the other hand decision trees are very suitable to ensemble methods like Random Forests which overcome some of these deficiencies. Due to its bagging mechanism and feature importance evaluation, Random Forests handle imbalanced datasets quite well, its robustness in overfitting reduction has improved generalization [5], [6]. Aggregation of different decision trees predictions helps Random Forests outperform other traditional models in the unfavourable circumstances previously mentioned, hence being preferable for fraud detection tasks.

C. Random Forests for Fraud Detection

Random Forests have gained recognition as an effective method for fraud detection due to their ability to handle high-dimensional data, identify complex patterns, and provide robust predictions. A major advantage of Random Forests lies in their capability to address imbalanced datasets, a common challenge in fraud detection, where fraudulent transactions often make up less than 1% of the data [5]. The ensemble nature of Random Forests, through bagging and random sampling of data and features, helps alleviate the issue of class imbalance by enabling the model to learn from diverse perspectives and improving its sensitivity to fraud [9].

In addition to handling imbalance, Random Forests offer valuable tools for feature selection and importance evaluation. Highlighted by the relevance of key variables in fraud detection [6]. By adjusting the random sampling during training, these variables can be prioritised, enhancing model accuracy. Random Forests use metrics like mean decrease in accuracy and GINI coefficient to assess variable importance, allowing for a better understanding of which features most significantly contribute to fraud detection.

Despite their strengths, Random Forests are not without limitations. One challenge is their computational cost, particularly with large datasets. Additionally, while feature importance provides some interpretability, the overall decision-making process of Random Forests remains somewhat opaque -yet far more comprehensible than that of deep neural networks- and can be referred to as a "black box" model. Further research is needed to improve the interpretability of these models and explore hybrid approaches that combine Random Forests with other techniques [10], [11].

Studies have demonstrated the effectiveness of Random Forests in fraud detection, with some reporting impressive performance, achieving a fraud detection accuracy of 99.6% using Random Forests, significantly outperforming other models like Naive Bayes and Nearest Neighbour [12]. However, the robustness of Random Forests can sometimes mask potential overfitting issues, especially when evaluated on single datasets [6].

Random Forests have also shown promise when combined with data balancing techniques like SMOTE, demonstrating that the combination of SMOTE and hyperparameter tuning improved Random Forests' accuracy and F1 scores, effectively addressing the challenges of imbalanced data [8].

However, the use of synthetic data requires caution, as it can sometimes lead to overfitting or reduced generalisability [7].

III. METHODOLOGY

A. Dataset

During testing, an attempt was made to evaluate multiple datasets sourced from Kaggle. However, during testing some of these datasets were proven to not be very sizable and limited evaluation. Some too small would be easily overfitted and result in a 100% across all metrics which was, others, were far too imbalanced with a non-fraud to fraud ratio of less than 0.3% paired with lacking feature or sample counts this induced too much bias into the model. The only dataset which we found suitable for our study was the IEEE-CIS fraud detection dataset [13].

The IEEE-CIS Fraud Detection dataset contains 590,540 transactions, of which approximately 3% are labelled as fraudulent, reflecting a significant class imbalance. It is divided into two components: transaction data and identity data, linked by a common TransactionID, few transactions were linked to identity data. The dataset includes both numerical features, such as TransactionAmt and TransactionDT, and categorical features, such as ProductCD, DeviceType, and P_emaildomain. Additionally, anonymised variables (V1 to V339) some of which were mostly null values contribute to the dataset's high dimensionality and sparsity. Preprocessing steps would need to be applied to address data inconsistencies and ensure the dataset was suitable for modelling.

B. Data Preprocessing

Preprocessing focused on cleaning the dataset, encoding features, and addressing the class imbalance. Missing numerical values were imputed using the median to handle outliers effectively, while categorical variables were imputed using the mode to maintain categorical structure. Categorical features in the form of labels were encoded using ordinal encoding in the order in which they were presented in the dataset, chosen for its computational efficiency despite potential risks of introducing spurious ordinal relationships. Severe class imbalance was mitigated using three techniques: SMOTE, random oversampling of the minority class, and random undersampling of the majority class. These preprocessing steps aimed to ensure the data was consistent, balanced, and encoded such that the Random Forest implementation could benefit from it.

C. Model Implementation

The Random Forest implementation of the python scikit-learn library was selected as the primary classification algorithm due to its robust documentation, support for parallel training, and large set of supporting libraries. The model could be configured with hyperparameters informed by literature and experimentation. A baseline model was decided upon with 100 decision trees (n_estimators) to balance computational efficiency and prediction accuracy; each tree's maximum depth limited to 10 to prevent overfitting, and the

size of the feature subset for splits (max_features) was set to the square root of the total number of features, promoting diversity among specialized trees.

D. Feature Selection

Feature selection was implemented using importance metrics, mainly mean decrease in accuracy and GINI coefficient, to identify the most influential variables. These metrics enabled the exclusion of less significant features, which enhanced model performance and reduced computational costs. This process ensured that the model concentrated on the most critical features, as suggested by previous research, particularly variables related to transaction and identity characteristics [11].

E. Model Refinement

Advanced techniques were incorporated to refine the model and improve its robustness. Bagging, inherent to Random Forests, aggregated feature subsets for individual trees to enhance overall accuracy and stability within a range of predictors. Boosting was also explored as an improvement method, focusing on correcting errors iteratively in successive trees. While boosting demonstrated higher recall, its computational demands and susceptibility to overfitting presented challenges. Hyperparameter tuning further optimised the model, ensuring effective utilisation of the dataset.

To evaluate the model's performance, weighted F1-score was used as primary metric, supported with fraudulent transaction specific F1-score. Given the dataset's imbalance, the evaluation prioritised minimising false negatives to improve fraud detection recall, recognising the critical importance of identifying fraudulent cases in real-world applications. By focusing on these metrics, the methodology provided a reliable framework for assessing and optimising the Random Forest classifier.

IV. EXPERIMENTAL RESULTS

A. Baseline Performance

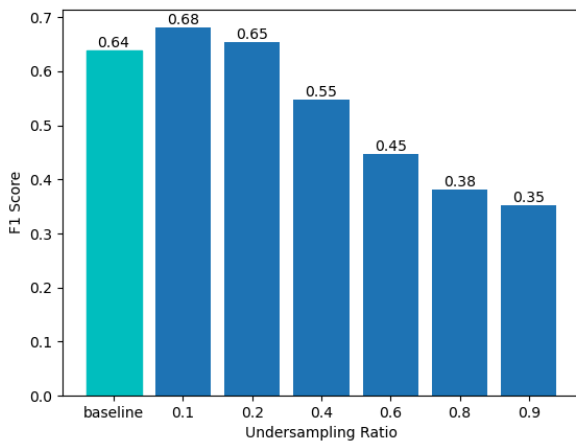
The baseline Random Forest model was evaluated on the imbalanced dataset without applying data balancing or refinement techniques. While the model achieved an overall accuracy of 98.7%, this primarily reflected the dominance of legitimate transactions rather than effective fraud detection. Metrics specific to fraudulent transactions indicated significant limitations: precision was 0.95, suggesting reasonable confidence in identified fraud cases, but recall was only 0.48, highlighting poor sensitivity. Consequently, the F1-score was 0.64, underscoring the model's inability to effectively detect fraudulent transactions in an imbalanced setting. These results emphasise the necessity of interventions to improve recall while preserving precision.

B. Impact of Data Balancing

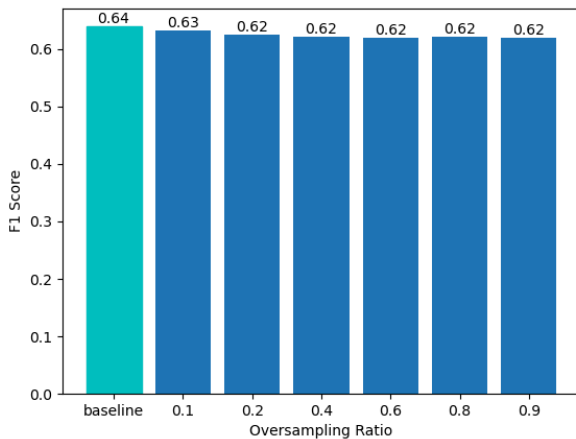
The application of data balancing techniques significantly influenced the performance of the Random Forest model in addressing the imbalance between legitimate and fraudulent transactions. Three balancing methods— random undersam-

pling and oversampling, and SMOTE—were evaluated to improve the recall of fraudulent cases, each demonstrating distinct trade-offs.

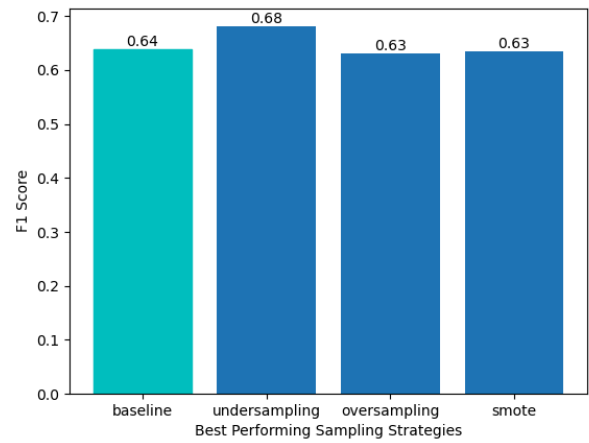
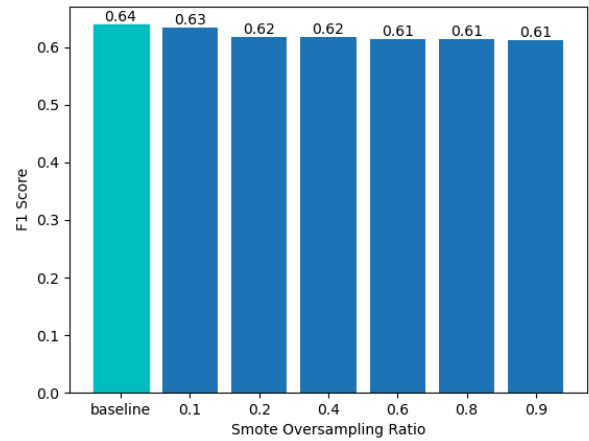
Undersampling to a ratio of 0.1 Fraud/NonFraud successfully boosted Fraud recall achieved the highest F1-score (0.68) among these methods. However, this came at the cost of significantly reduced precision which worsened with increased undersampling ratios (0.84 down to 0.22) presenting as a sharp decline in F1-Score, which led to a significant increase in false positives. This trade-off, while improving the detection of fraudulent cases, required careful undersampling ratio selection and raised concerns about operational feasibility due to the associated costs of investigating false alarms.



Oversampling demonstrated a more balanced but less favourable outcome, with a decreased recall of 0.47 and unchanged precision. This approach degraded model performance with a best F1-score of 0.63.



SMOTE failed to improve upon random oversampling, producing similar but worse metrics, likely due to the synthetic nature of the oversampled data.



C. Impact of Refinement Techniques

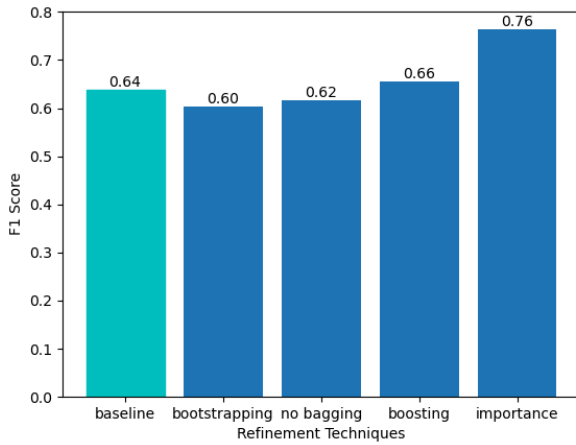
The effects on fraud detection performance for Ensemble Classifier of refinement techniques, including boosting, bagging, bootstrapping, and feature importance based reduction, were measured. Bagging was shown to be essential to the generalization ability of the Random Forest model, and feature importance based feature selection yielded an overall better performing model.

Bagging being an inherent feature of Random Forests, a degraded model with bagging disabled was trained, which increased fraud recall to 0.60 at the cost of greatly decreasing precision to 0.64. Without bagging the ability of the model to learn the minority class was hindered and sustained a marginally decreased F1-score of 0.62 due to unchanged majority class predictions. The high precision in the baseline model likely originated from the bagging mechanism. Bootstrapping however, yielded a less effective model with an F1-score of 0.60.

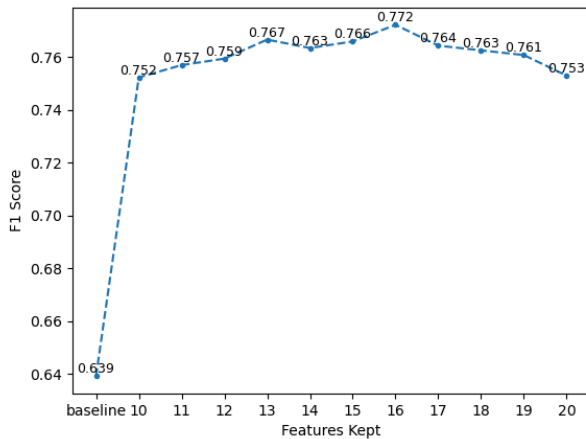
Boosting techniques significantly improved model ability to detect fraudulent transactions. XGBoost, an advanced boosting algorithm enhanced fraud recall to 0.51 without compromising precision which produced an F1-score of 0.66.

Feature reduction by dropping all but a subset of the most important features improved the F1-score to 0.77. The

result we found is highly related to the number of features remaining.

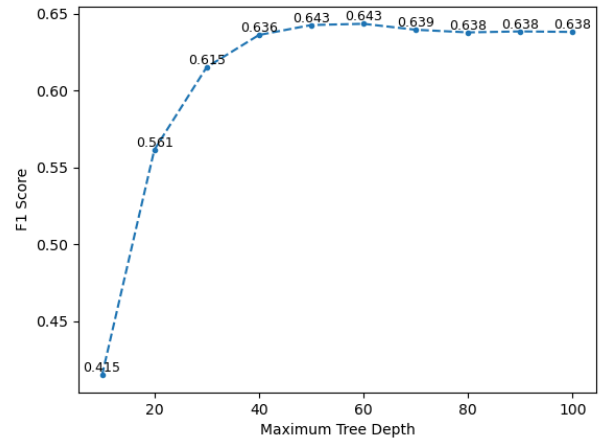


Given the sensitivity of the random forest and the individual trees to the combination of features which were trained on, and the large number of features in the dataset (433 columns), the GINI importance was used to determine a subset of features that contributed the most to the baseline model performance. A new model was trained on reduced feature sets of varying sizes achieving much improved F1-scores, a set of 16 features was found to be optimal achieving an F1-score of 0.77 with a fraud recall of 0.65. Despite the improvement, concerns can be raised about the reliability of this method given the implied loss of generalization power from dropped features and the tendency of random forest importance metrics to be misleading (GINI coefficient being susceptible to high cardinality features, and permutation importance being insufficient for datasets with too many features) [11].



D. Hyperparameter Tuning

Optimal parameter combination is highly dependent on the input data, but general trends can be observed from the improving fraud F1-scores presented by a search over the main random forest parameters. The most significant benefits came from the value of maximum decision tree depth

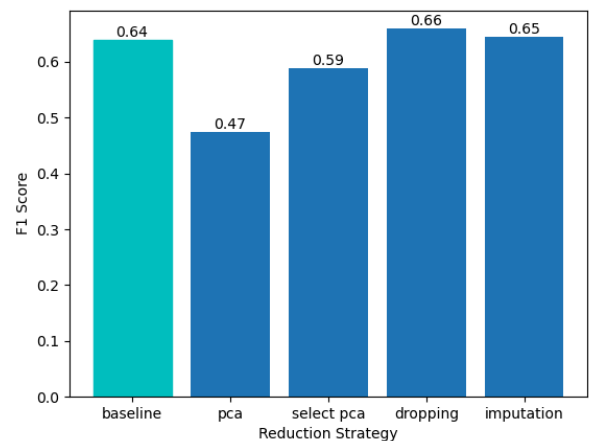


E. Preprocessing Methods

The preprocessing techniques employed in this study align closely with those reported in previous research. However, not consistent with earlier findings, dimensionality reduction through PCA greatly decreased model performance with a 0.47 F1-score, and failed to match baseline performance even when paired with less powerful feature selection using feature importance ranking to consolidate less significant features (F1-score of only 0.59); in contrast to the expected improvement implied by the noise reduction nature of the procedure that could reduce the number of trees needed to increase predictive power.

Handling missing values using median imputation for numerical features and mode imputation for categorical features presented minor model improvements.

Highlighting the success of importance based feature subset selection, also seen in previous works; dropping features that were composed mostly of null values improved model performance achieving an F1-score of 0.66 [6].



F. Model Refinement and Comparison with Results from Previous Works

The experimental results of this research were compared with findings from prior works on financial fraud detection using Random Forest and related techniques.

Boosting models, particularly XGBoost, demonstrated satisfactory performance compared to baseline Random Forest implementation. This aligns with previous studies that highlighted the enhanced recall capability of boosting techniques. In this study, XGBoost achieved an F1-score of 0.66 and a recall of 0.51, reaffirming its effectiveness in addressing class imbalance.

Our findings confirmed the modest impact of bagging without complementary strategies such as data balancing or feature engineering; bootstrapping presented no positive results. This suggests that while bagging and bootstrapping may enhance diversity in Random Forest models, their standalone contributions to fraud detection remain insufficient, and need to be supplemented with data balancing techniques.

V. DISCUSSION

A. Interpretation of Results

The findings of this study highlight the critical importance of data balancing and refinement techniques in enhancing machine learning models for financial fraud detection. Initial evaluation of the Random Forest model on an imbalanced dataset demonstrated significant limitations, particularly in fraud recall (0.48), which reflects the model's inability to detect fraudulent transactions effectively. This outcome underscores the challenge of class imbalance, where the model tends to favour the majority class, neglecting the minority class of fraudulent cases.

Data balancing techniques substantially improved the model's performance. Limited random undersampling emerged as the most effective method (0.68 F1-score), achieving a balanced trade-off between precision and recall without significantly increasing false positives; however this only sustained up to a threshold undersampling ratio. Oversampling techniques failed to deliver improving results, likely due to the synthetic and repetitive nature of the data which slightly reduced generalisation, this paired with the increased training cost for the expanded dataset rendered these methods impractical. All sampling techniques yielded degrading results at increased sampling ratios emphasising the importance of selecting a balancing method and thresholds that aligns with operational priorities, such as minimising undetected fraud (false negatives) and controlling the cost of false positives.

Refinement techniques significantly enhanced the model's ability to detect fraudulent transactions; feature subset selection based on importance metrics was particularly promising if used with caution such as to not lose generality. The effectiveness of feature subset selection was likely due to the sensitivity to feature count and distribution of the random forest, induced by the bagging mechanism, which relies on varied subsets of features to achieve robustness. Boosting

techniques also presented promising results, supporting the argument for their use in fraud detection where noise is often minimal.

Irregularity reduction methods such as PCA, column dropping, and imputation yielded worse or marginally improved models likely due to the near normalized nature of fraud datasets in which most transactions will likely be similar and attempting to regularise transactions could instead hide the indicators of fraudulent transactions.

B. Addressing Research Questions

- 1) How does data balancing affect Random Forest performance in financial fraud detection?

Data balancing techniques are essential for improving recall in fraud detection. Limited undersampling proved to be the most effective approach, offering balanced improvements in precision and recall. In contrast, oversampling techniques or undersampling excessively disturbed the data distribution degrading the model, making it impractical for operational use. These findings underline the importance of carefully selecting a balancing strategy that aligns with the specific goals of fraud detection systems.

- 2) How can Random Forest refinement methods like boosting, bagging, and bootstrapping improve model accuracy?

Refinement techniques significantly enhance model sensitivity to fraudulent transactions. Boosting methods, particularly XGBoost, demonstrated superior recall and F1-scores, solidifying the advantage of their use. In comparison, bagging and bootstrapping showed limited improvement, aligning with their less targeted approach to addressing class imbalance.

- 3) Can we replicate the successes of past research papers and pinpoint what made them successful?

This study replicated several key findings from prior research, particularly the effectiveness of boosting and feature selection in enhancing fraud detection performance. Feature importance rankings with hyperparameter tuning resulted in improvements over baseline comparable to those reported in previous studies, demonstrating consistency with earlier results and reaffirming the robustness of these techniques. The success of random forest refinement technique was found to depend on its synergy with the bagging mechanism which controlled how changes to the input data propagated to the training process of each decision tree.

These findings underscore the value of integrating class balancing, refinement methods, and preprocessing strategies to build robust and scalable fraud detection models, aligning model design with operational priorities ensures practical applicability in real-world financial systems.

C. Key Challenges

This study encountered several challenges, reflecting the complexities of developing machine learning models for financial fraud detection. One significant difficulty was hyperparameter tuning, particularly for the Random Forest and XGBoost models. While grid search cross-validation was effective in identifying optimal configurations, it was computationally intensive, especially when exploring combinations of parameters such as the number of estimators,

tree depth, and class weights. This trade-off between model optimisation and computational efficiency underscores the scalability constraints of hyperparameter tuning for large datasets, as noted in previous research [14].

Another major challenge was the high computational cost associated with training and evaluating models, particularly boosting algorithms like XGBoost. These models, while providing superior performance, demand substantial resources due to their iterative optimisation processes. This aligns with prior findings, which emphasise the balance required between performance and resource availability, particularly in large-scale fraud detection tasks [5].

Replicating results from prior studies posed additional difficulties due to dataset-specific nuances. Undersampling techniques, while generally effective, are likely to show varying results depending on the structure and prevalence of fraudulent transactions of the dataset. This highlights the importance of dataset-specific tuning and the challenges in generalising findings across different contexts.

Part of the initially proposed methodology for this study was to find similar but separate datasets on which to train and evaluate models such as to avoid overfitting and determine if good generalization was achieved. However, due to the often anonymized and highly preprocessed nature of financial fraud datasets, this was not possible.

Lastly, balancing operational priorities with model trade-offs would present a persistent challenge. Achieving an optimal balance between precision and recall to minimise both false positives and false negatives requires careful selection of data balancing and refinement techniques. This issue mirrors a broader concern in fraud detection research, where operational constraints often influence the practical feasibility of deploying advanced models [15].

Addressing these challenges involved iterative experimentation and significant computational resources. The findings underscore the importance of scalability, reproducibility, and adaptability in the design and implementation of machine learning models for financial fraud detection.

D. Implications for Fraud Detection

The findings of this study provide critical insights into the design and implementation of fraud detection systems in financial institutions. The demonstrated effectiveness of feature selection and boosting techniques, particularly XGBoost, underscores their potential for deployment in systems where detecting rare events, such as fraudulent transactions, is paramount. These techniques balance precision and recall, minimising false negatives (undetected fraud) while maintaining operationally acceptable levels of false positives (legitimate transactions flagged as fraud). This balance is essential for financial institutions to manage resource allocation effectively and maintain customer trust.

From an operational perspective, boosting techniques for Random Forest offers a robust solution for addressing class imbalances without generating synthetic data or compromising precision excessively. Compared to methods like undersampling or SMOTE, boosting aligns well with institutional priorities for minimising disruptions to legitimate

transactions while maintaining high sensitivity to fraudulent cases.

The methodologies developed in this research extend beyond fraud detection to broader anomaly detection domains, including cybersecurity, healthcare, and industrial monitoring. Random Forest's ability to handle high-dimensional data and imbalanced classes makes it well-suited for detecting rare events such as network intrusions, disease outbreaks, or equipment failures. For example, feature selection and class balancing methods from this study can be adapted to identify anomalies in highly imbalanced datasets.

Boosting techniques, particularly XGBoost, further enhance applicability by providing higher recall and precision in detecting subtle patterns within complex datasets. This adaptability ensures these methods can be fine-tuned for specific anomaly detection tasks, maintaining high accuracy and scalability across domains.

In summary, this research advances fraud detection methodologies while highlighting the broader applicability of Random Forest and boosting techniques to anomaly detection in diverse fields. By addressing challenges such as class imbalance, feature selection, and model optimisation, the proposed approaches provide a versatile framework for identifying rare and impactful events in real-world settings.

VI. CONCLUSION

A. Summary of Findings

This study explored the effectiveness of significant advancements in fraud detection using Random Forest models enhanced with current data balancing and refinement techniques. The baseline model, despite high precision (0.95), exhibited low recall (0.48), underscoring its limitations in detecting minority-class instances. Carefully incorporating class balancing methods, such as undersampling, substantially improved recall, with feature selection based on importance metrics providing the most balanced performance (recall: 0.65; F1-score: 0.77). Boosting techniques, particularly XGBoost, improved upon the Random Forest baseline, achieving a recall of 0.51 and an F1-score of 0.66. These findings affirm the importance of preprocessing, feature reduction, and refinement strategies in addressing class imbalance and improving fraud detection. The integration of data preprocessing methods, such as PCA and imputation delivered less promising results. By systematically assessing balancing and refinement strategies, this study bridges the gap between academic methodologies and operational requirements, offering actionable insights for developing robust fraud detection systems.

B. Future Work

Future research should evaluate the generalisability of these findings across diverse datasets and investigate advanced hyperparameter optimisation techniques, such as Bayesian optimisation, to improve performance while reducing computational costs. Comparative studies involving state-of-the-art deep learning models, could offer insights into their potential advantages over Random Forests and boosting

methods. Further improvements could also be achieved with more refined data preprocessing methods. Additionally, incorporating domain-specific knowledge into feature engineering may further enhance model accuracy and applicability across various anomaly detection tasks beyond financial fraud.

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